

Question				Marking details	Marks Available					
					AO1	AO2	AO3	Total	Maths	Prac
4 HT	(a)	(i)		Time taken for a halving of (1) number of (radioactive) nuclei / mass / atoms/ amount of material / amount of substance/activity (1)	2			2		
		(ii)		4.2 billion = 4.2×10^9 (1) number of half lives = $4.2 \times \frac{10^9}{7} \times 10^8 = 6$ (1) fraction = $\frac{1}{2^n} = \frac{1}{64}$ (1)		3		3	3	
	(b)	(i)		Control rods (1) Absorb (excess) neutrons (1) Allowing one fission neutron to continue (to cause more fission) (1)	3			3		
		(ii)		Contains 56 protons (1) 85 neutrons (1)		2		2		
	(c)			$\begin{array}{c} {}_{92}^{239}\text{U} \text{ (1)} \\ \\ \rightarrow {}_{93}^{239}\text{Np} \text{ (1)} + {}_{-1}^0\beta \\ \\ \rightarrow {}_{94}^{239}\text{Pu} \text{ (1)} + {}_{-1}^0\beta \\ \\ \text{both beta (1)} \end{array}$		4		4		
				Question 4 total 14	5	9	0	14	3	0